Francisco Ramirez Serrano

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Summary

Robotics engineer and first author in *Science Robotics* who has taken robots from first prototype to working system, from insect-scale jumping robots to a humanoid running a vision-language-action (VLA) AI model. Strongest when testing, judging and explaining systems to the people who rely on them. Seeking an in-person role in robot evaluation, safety and compliance, integration or deployment.

Experience

Graduate Researcher – Humanoid Robots & Responsible AI Sept. 2025 – Present
MIT Computer Science and Artificial Intelligence Laboratory (CSAIL), Living Lab *Cambridge, MA*

- Brought up a Unitree G1 humanoid with dexterous hands from scratch: integrated and deployed NVIDIA's GR00T VLA foundation model, including teleoperation, camera streaming and robot networking (featured in a CSAIL video).
- Audited the deployed model's safety and ethics (stakeholder workshop, deployment audit); explored entrepreneurship opportunities through E14 Beyond the Lab and MIT Fuse.

Teaching Assistant – 6.UAR, Preparation for Undergraduate Research Spring 2026, Fall 2026
MIT Department of Electrical Engineering and Computer Science *Cambridge, MA*

- Evaluate research proposals and writing for 30+ MIT students per term and meet each four times to review project progress.

Product Design Engineer (Consultant) – Consumer Electronics June – Sept. 2024
Nothing Technology *London, UK*

- Developed new test methods to evaluate the ergonomics of the Headphone (1) (launched 2025), and turned force-measurement data into design and manufacturing recommendations.

Research Fellow – Impulsive Robotics & Surgical Devices Nov. 2021 – Jan. 2024
Harvard Microrobotics Laboratory (Prof. Robert Wood) *Cambridge, MA*

- Independently led a walking-jumping microrobot project that earlier attempts had not achieved, from concept to first-author papers in *Science Robotics* and *Advanced Materials Technologies*.
- At Project 1985, a surgical-robotics start-up launched from the lab (2022–23): designed and prototyped hydraulically actuated millimeter-scale surgical tools; joined strategy discussions with venture investors.

Research & Development Engineer – Drone Aerodynamics May – Oct. 2021
Flybotix, indoor inspection drone start-up *Lausanne, Switzerland*

- Led design and build of a test bench for co-axial propeller configurations, cutting bench costs by 75%.

Education

PhD Student, Electrical Engineering and Computer Science Sept. 2024 – Present
Massachusetts Institute of Technology (MIT) *Cambridge, MA*

- Available for full-time roles from early 2027.
- MIT SERC Scholar (Social and Ethical Responsibilities of Computing), 2025–26. Shenzhen Scalable HCI residency (Jan. 2025): won a hardware hackathon, studied factory production flows.

Master of Science, Robotics Sept. 2019 – Oct. 2022
EPFL (Swiss Federal Institute of Technology Lausanne); thesis at Harvard University *Lausanne, Switzerland & Cambridge, MA*

Bachelor of Science, Microengineering Sept. 2016 – July 2019
EPFL; exchange year at the Technical University of Munich (TUM) *Lausanne, Switzerland & Munich, Germany*

Publications, Presentations & Peer Review

F. Ramirez Serrano, …, R. J. Wood. "A springtail-inspired multi-modal walking-jumping microrobot." *Science Robotics*, 2025.
F. Ramirez Serrano et al. "Considerations for the design and rapid manufacturing of pop-up MEMS devices." *Adv. Mater. Technol.*, 2024.
C. Vourtsis, V. C. Rochel, **F. Ramirez Serrano**, …, D. Floreano. "Insect inspired self-righting for fixed-wing drones." *IEEE RA-L*, 2021.
Presentations: ICRA 2022 workshop (Philadelphia); posters at Scalable HCI, Shenzhen (2025) and MIT SERC Symposium (2026).
Peer reviewer for *IEEE Robotics and Automation Letters* (RA-L), 2023.

Leadership, Teaching & Activities

Teaching: robotics tutor, TechLabs Geneva (2024); children's AI workshop, Lithuania (2025); guest teacher, 'Iolani School (2023).
Community: Social Chair, Spain@MIT; volunteer wingfoil instructor, MIT Sailing; intramural soccer champion.

Skills

Languages: Spanish (native), English & French (fluent), German & Italian (proficient).
Have lived in: Costa Rica, Senegal, Benin, Italy, Germany, Switzerland, the UK and the US.
Robotics & AI: VLA deployment (NVIDIA GR00T), humanoid teleoperation, ROS, Python, C, MATLAB, Linux, GPU clusters, simulation.
Testing & hardware: test-rig design, high-speed video, force/torque sensing, motion capture, design for manufacturing.